SOPCOS

Industrial Trust Protocol

Machines Execute. Protocol Adjudicates. Ledger Remembers.

Authority: Protocol

Terminal Walkthrough

NotebookLM

01_Reset_and_Start_Ledger.ps1

Reset Ledger and prepare workspace

Recompile Core Ledger

Initialize Genesis and Validator nodes

Bootstrap network state

```
[1/5] Purging badgerDB...
Cleaning: badger_3000

[3/5] Building core-ledger binary...
✅ Binary compiled successfully.

Stabilizing Genesis Node (3s)...

--- MULTI-NODE STARTUP COMPLETE ---
```

NotebookLM

02_Account_and_Finance_Setup.ps1

Provision new decentralized identities

Prepare Operator and System accounts

Execute test funding transfers

Initialize network economic structure

```
→ Registering Root Organization: did:sop:acorp
→ Registering Edge Gateway: did:sop:acorp-synapse-edge-01
→ Registering Operator: did:sop:acorp-operator-01
→ Registering Auditor: did:sop:acorp-auditor-01
→ Registering Auditor: did:sop:acorp-auditor-01
→ Registering Edge Gateway: did:sop:acorp-synapser-01
→ Registering Edge Gateway: did:sop:acorp-oonator-01

Executing Operational Funding (100 SOPC each)...
✅ Accounts created successfully.
```

NotebookLM

03_Deploy_Smart_Unit.ps1

- Prepare WASM-based Smart Unit
- Create identity and asset records (IDAS)
- Deploy Smart Unit to the Ledger
- Activate operational environment

```
Compiling Smart Unit WASM (TinyGo)...
registry.wasm compiled successfully.

Creating IDAS Asset Class (Industrial Boilers)...

Calculated ClassID: 16372392958147715323

Mint IDAS transaction sent successfully.
OP_MINT_ASSET -> did:sop:boiler-01
```

NotebookLM

04_Axon_Policy_Deployment.ps1

- Compile deterministic policy
- Upload policy artifact to Cloud Vault
- Anchor cryptographic hash to Ledger
- Finalize policy enforcement mechanism

```
Publishing WASM to Cloud Vault (S3)...
URN: urn:sopcos:policy:278a9ac890b498c5
content_hash:
278a9ac890b498c5e95966d7a2efcd228f1477fc9016c
9a21f6eb3e8fdf4863a0d
OP_ARTIFACT_REG Broadcast...
✅ Policy anchored to Ledger successfully.
```

NotebookLM

05_Launch_Synapse_Runtime.ps1

Start Axon L2 Gateway service

Ignite Synapse deterministic runtime

Activate live telemetry stream

Engage Hot-Path decision engine

```
[DEBUG] Raw Telemetry Payload: {"device_id":"did:sop:b
oiler-01","pressurerature":188.64,"vibration":0.019}
[SMART UNIT] Device: did:sop:boiler-01 | Verdict: ALL
ALL_SYSTEMS_NOMINAL
[DEBUG] Raw Telemetry Payload:
[SMART UNIT] Device: did:sop:boiler-01 | Verdict: ALLOW
ALL_SYSTEMS_NOMINAL
[DEBUG] Raw Telemetry Payload: {"device_id":"did:sop:
boiler-01","pressure":95.08,"temperature":231.34,"vibr
ation":0.025}

[SMART UNIT] Device: did:sop:boiler-01 | Verdict: WARN
| Rule: BOILER_TEMPERATURE_EXCEEDS_NORMAL_ENVELOPE
```

NotebookLM

06_vinci_test.ps1

- Test human operator intervention
- Execute Emergency Override scenario
- Observe 'DIRTY STATE' transition
- Verify protocol authority and liability transfer

```
Operator Identity Loaded: did:sop:acorp-operator-01
Operator Identity Loaded: did:sop:acorp-operator-01
OP_CONFESSION Broadcast...
 justification_code: CLASS_C
 reason: Manual Override due to sensor failure
 liability_hash: c2hhMjU2OmR1bW15X3BvbGljeV9oYXNo

BROADCAST SUCCESS: The System is now in DIRTY STATE!
Tx_ID: 7e3afcacf6f14c7d88114a76d0f957c66c682081
```

NotebookLM

07_auditor_reset_test.ps1

Initiate Level 5 Auditor reset process

Restore system to ‘CLEAN STATE’

Verify immutable audit trail

Complete forensic liability attribution

```
Auditor Identity Loaded. Target: did:sop:boiler-01
Requesting Reset Template (Type 18)...
OP:STATE_RESET -> auditor_verdict: JUSTIFIED
Reason: Compliance Audit #9921 Passed. System restored.
Tx_ID: d8717454875297ff8a6849ac54431b6e0070151de
Local State for did:sop:boiler-01 should now be CLEAN.
```

NotebookLM

SOPCOS

Industrial Trust Requires Protocol Authority

Machines Execute. Protocol Adjudicates. Ledger Remembers.

Authority: Protocol.

NotebookLM